PAYLOAD DELIVERY MISSION

INTRODUCTION

The Payload Delivery Mission simulates a real-world drone logistics challenge. Teams must navigate obstacles, transport a payload, and deliver it accurately to the target zone. The mission tests precision flying, stability under load, and safe landing capabilities. It reflects practical applications such as medical supply and disaster relief delivery.

PROBLEM STATEMENT

Design and operate a remotely controlled drone system capable of executing a complete payload delivery mission within a structured obstacle environment. The drone must be manually piloted through a predefined course that replicates real-world challenges such as confined spaces and physical barriers, while maintaining flight stability, thrust balance, and structural reliability under additional payload load. It should ensure safe and secure transportation of the payload without compromising maneuverability or control.

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The drone must demonstrate precise manual navigation and effective obstacle avoidance throughout the mission, maintaining smooth and stable flight. Upon reaching the designated target area, the payload must be released accurately using a controlled deployment mechanism. Following delivery, the drone must be guided to a specified landing zone and execute a safe, controlled landing.

The problem statement reflects practical real-world applications such as emergency response, medical supply transport, disaster relief operations, and last-mile delivery, with an emphasis on pilot skill, reliability, precision, and operational efficiency.

DRONE SPECIFICATION

Drone Specifications

- Recommended flight time: 20 minutes (not a strict condition; no marks deduction)
- Frame size: 300-550 frame, max 10 inch propeller
- Weight < 3kg.
- It should have a strong receiver and transmitter to operate in small spaces with walls and other structures that can potentially block signals.
- Battery Voltage <= 24 V

PAYLOAD SPECIFICATION

- Weight : Approximately 200 - 500 grams (exact value will be specified on the day of the event)
- Shape : The payload may be of any shape, size, or configuration

ABSTRACT SUBMISSION ROUND

The participating teams must submit an abstract of their drone's concept, in proper documented form. This should include its configuration, design, features, estimated cost, and an approach to tackle the problem statement. The teams must compile all this information in a PDF file. If any extra files are to be included, the submission must be in the form of a ZIP file. Please note that the files submitted should NOT be password-protected or corrupted. In either of the cases, the submissions would be directly rejected.

GUIDELINES FOR ABSTRACT SUBMISSION

Rough sketches of your idea including the Design Report of your model. The abstract must specify the use of software and the mechanism used in the design analysis of the drone.

- USPs (Unique Selling Propositions) of your design and innovations made by you should be mentioned in the report and mention how they are going to tackle the specific challenges.
- An abstract of the idea needs to be uploaded on the Dronotics website before the deadline.
- The abstract should not exceed 500 words and should include the following details of all team members along with the Team.

Cover Page Details:

The cover page of the submission must include the following information:

- Team Name
- Team Leader Details:
  - Name of Team Leader
  - College / Institute Name
  - Year of Study
  - Email ID
  - Phone Number
- Team Members:
  - Names of all team members

File Naming Convention:

The file must be submitted in the following format:

PDM_<TeamName>_<LeaderName>_abstract.pdf

Example (for team ABC): PDM_ABC_AmanGandotra_abstract.pdf

Submission Email

The abstract/design report must be mailed to:
dronoticsjiit128@gmail.com

The Subject of email should be in this format:
Payload Delivery Mission - <TeamName> - <LeaderName> Abstract
Example: Payload Delivery Mission - ABC - Aman Gandotra Abstract

Important Note

- Only PDF format submissions will be accepted.
- Files must not be corrupted or password-protected.
- Any submission not following the naming convention may be rejected.

ROUND 1

- Round 1 is designed to evaluate the drone's lifting capability, stability under load, and controlled payload delivery under manual operation.
- The payload will be pre-attached to the drone before the start of the run
- The drone must perform a manual takeoff from the designated starting zone
- After takeoff, the drone must:
  - Lift the payload to a specified height(will be specified on day of event)
  - Navigate to the designated target drop zone
  - Stabilize and hover above the target area for 3-5 seconds
  - Execute a controlled payload drop within the marked zone

- The drone must maintain stable flight while carrying the payload
- Payload release must occur only when positioned over the target zone
- The drop should be controlled and precise, avoiding excessive impact

Evaluation Metrics:-

- Successful payload handling
- Navigation efficiency
- Delivery precision
- Return capability
- Landing quality
- Overall system reliability and stability

All metrics have not been disclosed and will be specified on day of event.

ROUND 2

- Round 2 is designed to evaluate the drone's advanced maneuverability, payload handling capability, and operational precision in a dynamic environment. The mission must be completed with controlled flight, efficient navigation, and accurate payload delivery.
- The drone will be placed on a designated starting platform
- Participants must initiate a manual takeoff to begin the mission
- The mission will consist of the following sequence:
  - Obstacle Navigation (Phase 1) : After takeoff, the drone must navigate through a series of obstacles placed at variable heights. The drone must maintain stable and controlled flight while clearing all obstacles and will reach payload station.
  - Payload Pickup : Upon reaching the payload station, the drone must secure the payload using one of the following methods:
    - Manual loading : Drone lands or descends, and the payload is attached manually
    - Autonomous pickup : Drone uses mechanisms such as grippers or hooks to pick the payload without human assistance
  - Obstacle Navigation (Phase 2) : After securing the payload, the drone must again navigate through obstacles while maintaining stability under load
  - Payload Delivery : The drone must reach the designated drop zone. It must hover at a height of approximately 3 meters for at least 5 seconds. After stabilizing, the drone must execute a controlled payload drop within the target area.
  - Return & Landing : After releasing the payload, the drone must return to the original takeoff platform and needs to execute a safe and controlled landing at the starting position

Evaluation Metrics:-

- Successful payload handling
- Navigation efficiency
- Delivery precision
- Return capability
- Landing quality
- Overall system reliability and stability

All metrics have not been disclosed and will be specified on day of event.

SCORING CRITERIA

Round 1 will be of 100 Marks.
Successful mission completion will be 20 marks.
Round 2 will also be of 100 marks.
(100 marks for round 2 will be distributed equally for completing mission and drone stability & control)
In case teams score equal points **completion time** will be considered for resolving.
The final decision lies in the hands of jury.

GUIDELINES

Eligibility: School students, undergraduates, graduates and working professionals are allowed in this event.
Team Size: A team can have a minimum of 3 and a maximum of 5 members.
No Double Troubles: Only one entry would be acceptable. In case of multiple entries, the latest one would be considered for evaluation.